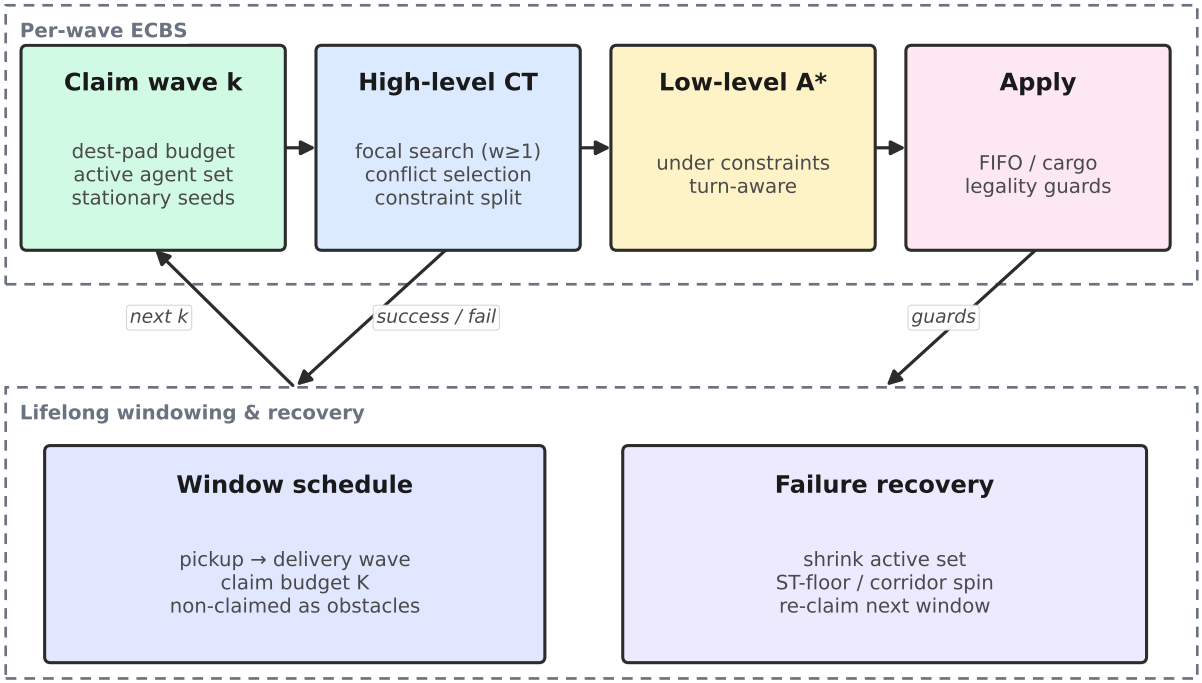Windowed ECBS Lifelong Wave Pipeline